

**MHD BOUNDARY LAYER FLOW OF MAXWELL FLUID OVER A
STRETCHING SURFACE WITH CONVECTIVE BOUNDARY
CONDITION IN THE PRESENCE OF CHEMICAL REACTION AND
NANOPARTICLES : A NUMERICAL STUDY**

**S. SUNEETHA
K.SUBBA RAYUDU
L.WAHIDUNNISA
&
B.MAMATHA**

Department of Applied Mathematics,
Yogi Vemana University, Kadapa-516003, INDIA
Department of Mathematics,
Sri Padmavathi Womens Degree and PG College, Tirupati, A.P.

suneethayvu@gmail.com

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ABSTRACT

Keyword:

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fluid , Boundary layer
flow, Chemical reaction,
heat
generation/absorption.

In this article, the influences of the convective boundary condition and heat generation/absorption on Magnetohydrodynamic boundary layer flow of a Maxwell fluid over a stretching surface in the presence of nanoparticles have been numerically investigated. In the model, the physical mechanisms responsible for Brownian motion and thermophoresis with a chemical reaction are considered. The resultant non-linear coupled ordinary differential equations are solved numerically by utilizing the fourth order Runge-Kutta method with shooting technique. Computational results are gotten for the velocity, temperature, the skin-friction coefficient, the rate of heat transfer and rate of mass transfer in the sheet. The evaluated results are compared with another analytical study reported earlier in scientific literatures. The rates of heat and mass transfer are enhanced with a destructive chemical reaction and Biot number.

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INTRODUCTION:

In our daily life we come across many non Newtonian fluids such as tomato ketchup, shampoo, sugar solutions, paints, certain oils, lubricants and so on. The analysis of non-Newtonian fluids is noteworthy because of various industrial and engineering applications. Such fluids are encountered in the process of manufacturing coated sheets, foods, drilling muds, cosmetic products, dilute polymer solutions, polymeric melts etc. For instance, if one uses a non- Newtonian fluid as the coolant or heat exchangers, the required pumping power may be substantially reduced. Therefore, the fundamental analysis of the flow field of non-Newtonian fluids in a boundary layer is a vital part in the study of fluid dynamics and heat transfer.

Study of non-Newtonian fluids over a stretching surface achieved great attention due to its large number of applications in engineering and industry, such as for chemicals, petrochemicals, molten plastics, electronic chips, bioengineering, and pharmaceuticals. Infact, the effects of non-Newtonian behaviour can be determined due to its elasticity, but sometimes rheological properties of fluid are identified by their constitutive equations. In view of rheological parameters, the constitutive equations in the non-Newtonian fluids are more complex and thus give rise the equations which are complicated than the Navier– Stokes equations. Crane [1] was the first who describe the analytical solution for a stretching sheet in laminar boundary layer flow. Some of the reviews of non-Newtonian fluid flow problems have been presented [2-4].

In recent years, research on the boundary layer flow problem with a convective surface boundary condition have gained much interest among researchers, since first introduced by Aziz [5], who considered the thermal boundary layer flow over a flat plate in a uniform free stream with a convective surface boundary condition. Ishak [6] obtained the similarity solutions for the steady laminar boundary layer flow over a permeable plate with a convective boundary condition. Makinde and Aziz [7] investigated numerically the effect of a convective boundary condition on the two dimensional boundary layer flows past a stretching sheet in a nano fluid.

Patil et al.,[8] studied the influence of convective boundary condition on double diffusive mixed convection from a permeable vertical surface.

A great deal of interest has been created in studying magnetohydrodynamic flows and heat transfer due to the effect of magnetic fields on the boundary layer flow control and on the performance of various systems which involve electrically conductive flows. In addition, this type of flow finds applications in many engineering problems such as MHD generators, Plasma studies, Nuclear reactors, and Geothermal energy extractions. Mustafa et al., [9] investigated the MHD Boundary Layer Flow of Second-Grade Nanofluid over a Stretching Sheet with Convective Boundary Conditions. Moreover, it is observed there is an appreciable increase in the temperature and the thermal boundary thickness when the strength of Brownian motion and thermophoresis effects are increased. Ibrahim & Haq [10] have numerically investigated the Magnetohydrodynamic (MHD) stagnation point flow of nanofluid past a stretching sheet with convective boundary condition.

The thermal conductivity of fluids plays an important role in the heat transfer. However, it is observed that enhancement of the thermal conductivity of poor heat transfer fluids is possible in view of the addition of nanoparticles in the base fluids. The nano particles can be found in metals such as (Cu, Ag), oxides (Al_2O_3), carbides (SiC), nitrides (AlN, SiN) or nonmetals (graphite, carbon nanotubes). Nanofluids have novel properties that make them potentially useful in many applications in heat transfer including microelectronics, fuel cells, pharmaceutical processes and hybrid-powered engines. Nanoparticles afford a bridge between bulk materials and molecular structure. The term “nanofluid” was first introduced by Choi [11]. The thermal conductivity of nanofluids depends on some factors such as particle size, volume fraction of particles in the suspension and intrinsic thermal conductivities of the base fluid and particles. Maxwell was one of the first to explore conduction properties analytically for a mixture consisting of particles embedded in a base medium following the effective medium theory. (Effective Medium Theory or Effective Medium Approximation denoted as EMT or EMA). The effective medium theory describes the macroscopic properties of a medium based on the properties of the components and their relative volume fractions.

The Maxwell model is one of the first models which predict the thermal conductivity of nano fluids. To account for these factors, the traditional Maxwell model for the effective thermal conductivity of a mixture is modified to report for nonlocal heat transport that can arise due to the small characteristic length in a nanofluid – the particle size thus naturally appears in the modified model. The resulting model is calibrated and validated with experimental measurements of alumina nanofluids: overall, good agreement is found. Furthermore, the modified model is shown to be consistent in the limits of only the base liquid as well as for large particle sizes. Nadeem et al. [12] investigated the numerical study of MHD boundary layer flow of a Maxwell fluid past a stretching sheet in the presence of nanoparticles. Akbar et al. [13] analyzed the numerical solutions of Magneto hydrodynamic boundary layer flow of tangent hyperbolic fluid towards a stretching sheet.

In addition, heat and mass transfer with chemical reaction has special significance in chemical and hydrometallurgical industries. The formation of smog represents a first order homogeneous chemical reaction. For instance, one can take into account the emission of NO₂ from automobiles and other smoke-stacks. Thus, NO₂ reacts chemically in the atmosphere with unburned hydrocarbons (aided by sunlight) and produces peroxyacetyl nitrate, which forms a layer of photochemical smog. Chemical reactions can be treated as either homogeneous or heterogeneous processes. It depends on whether they occur at an interface or as a single-phase volume reaction. Most of the Chemical reactions involve either heterogeneous or homogeneous processes. Some reactions are very slow or not at all, except in the presence of catalyst. A complex interaction lies between the homogeneous and heterogeneous reactions which is incorporated in the production and consumption of reactant species at different rates on the fluid and also on the catalytic surfaces, such are happened in fog formation and dispersion, food processing, manufacturing of polymer production, groves of fruit trees, moisture over agricultural fields, equipment design by chemical processing, crops damage via freezing etc. Some kind of chemical reaction is observed if there is a foreign mass in air or water. During a chemical reaction heat is generated between two species (Byron Bird R. et al. [14]).

Generally, the reaction rate depends on the concentration of the mass itself. If the rate of reaction is directly proportional to concentration itself (Cussler [15]), then it is said to be first order.

Chemical reactions are divided into two categories; via homogeneous reaction, which involves only single phase reaction and heterogeneous reaction, which involves two or more phases and occur at the interface between solid and fluid or between two fluids separated by an interface. The important applications of homogenous reactions are the combination of common household gas and oxygen to produce a flame and the reactions among aqueous solutions of acids and bases. Themelis [16] stated that the majority of chemical reactions experience in applications are first-order and heterogeneous reactions for instance, hydrolysis of methyl acetate in the presence of mineral acids and inversion of cane sugar in the presence of mineral acids. A chemical reaction is said to be first-order when a reaction rate depends on a single substance and the value of the exponent is one. Midya [17] observed from their study that the first-order chemical reaction is very significant in chemical engineering where the chemical reactions take place between a foreign mass and the working fluid. Krishnamurthy et al., [18] studied the effect of chemical reaction on MHD boundary layer flow and melting heat transfer of Williamson nanofluid in porous medium.

The effect of heat source/sink on heat transfer is another significant aspect in view of many physical problems. Heat generation or absorption may change the heat distribution in the fluid which consequently affects the particle deposition rate in the system such as semiconductors, electronic devices and nuclear reactors. Heat source/sink may be considered constant, space dependent or temperature dependent. Here, we will discuss non-uniform, i.e., space and temperature dependent heat source/sink. Hady et. al., [19] numerically investigated the effect of heat generation/absorption on natural convective boundary-layer flow from a vertical cone embedded in a porous medium filled with a non-Newtonian nanofluid. Ramesh and Gireesha [20] studied the influence of heat source/sink on a Maxwell fluid over a stretching surface with convective boundary condition in the presence of nanoparticles. A few investigations are reported by some researchers [21-22]. Numerically [19] investigated the effect of heat generation/absorption on natural convective boundary-

layer flow from a vertical cone embedded in a porous medium filled with a non-Newtonian nanofluid. Ramesh and Gireesha [20] studied the influence of heat source/sink on a Maxwell fluid over a stretching surface with convective boundary condition in the presence of nanoparticles. . A few investigations are reported by some researchers [21-22].

Generally, a viscous fluid normally sticks to a boundary, i.e., there is no slip of the fluid relative to the boundary. The no slip boundary state, which is the supposition that a liquid stick to a solid boundary, is one of the pillar of the Navier–Stokes theory. The non-adherence of the fluid to a solid boundary is called velocity slip. Velocity slip may arise on a stretching boundary when the fluid contains particulates such as emulsions, suspensions, foams and polymer solutions. Fluids exhibiting boundary slip also feature in diverse areas such as polishing of artificial heart valves and hydrophobic walls in fuel cells. Several studies on slip effects and some other related problems were performed by some investigators [23-25].

To the best of our knowledge, the influences of chemical reaction, convective boundary condition, and heat generation/absorption on MHD flow and the heat transfer characteristics of Maxwell nanofluids over a stretching surface with velocity slip have not been investigated. The governing partial differential equations have been transformed into a system of ordinary differential equations by similarity transformations which are then solved numerically by using fourth order Runge–Kutta scheme together with the shooting method. In this study, numerical results of the velocity, temperature, and nanoparticle concentration distributions are presented. The skin friction coefficient, local Nusselt number, and local Sherwood number are also presented and discussed.

Nomenclature

a	positive constant (s^{-1})
B_0	constant magnetic field($kg/(s^2A)$)
Bi	Biot number
C	nanoparticle concentration(kg/m^3)
C_∞, C_w	free stream and surface particle concentrations (kg/m^3)
C_f	skin friction coefficient
C_p	specific heat ($J/(kg K)$)

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D_B	Brownian diffusion coefficient (m^2/s)
D_T	thermophoretic diffusion coefficient (m^2/s)
$f(\eta)$	dimensionless free stream function
k	thermal conductivity of the fluid ($W/(m\ K)$)
Le	Lewis number
M	magnetic parameter
Nb	Brownian motion parameter
Nt	thermophoresis parameter
Nu_x	local Nusselt number
Sh_x	local Sherwood number
Pr	Prandtl number
T	fluid temperature (K)
T_∞, T_f	free stream and surface fluid temperatures (K)
u, v	velocity components along the x and y axes (m/s)
x, y	Cartesian coordinates (m)

Greek symbols

α	thermal diffusivity of the base fluid (m^2/s)
γ	chemical reaction parameter
δ	heat generation/absorption parameter
η	similarity independent variable
θ	dimensionless temperature
λ	elastic parameter
μ	Dynamic viscosity of the fluid ($kg/(m\ s)$)
ν	kinematic viscosity capacity of the nanoparticle material (m^2/s)
ρ_f	fluid density (kg/m^3)
$(\rho C)_f$	heat capacity of the fluid ($J/(m^3\ K)$)
$(\rho C)_p$	effective heat capacity of the nanoparticle material ($J/(m^3\ K)$)
σ	electrical conductivity (S/m)
τ	nanoparticle to base fluid heat capacity ratio
ϕ	dimensionless nanoparticle concentration
ψ	dimensionless stream function

Subscripts

w	surface condition
∞	Condition far away from the surface

MATHEMATICAL FORMULATION:

We consider a steady, laminar boundary layer of a non-Newtonian nanofluid past a stretching surface coinciding with the plane $y = 0$. The flow is confined to the region $y > 0$, where y is the coordinate measured normal to the stretching surface, as shown in Fig. 1. It is also assumed that the bottom surface of the stretching plate is heated by convection from a hot fluid with temperature T_f , which provides a heat transfer coefficient h_f . In this study, a first-order homogeneous chemical reaction of species with velocity slip is considered. The plate is stretched along x-axis with linear velocity

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$u_w = ax$, where a is a positive constant. A magnetic field with uniform strength B_0 is applied in the y direction (i.e., normal to the flow direction). The magnetic Reynolds number is assumed to be small. Hence, the induced magnetic field is small compared with the externally applied magnetic field. Under the above assumptions, the boundary layer equations of a Maxwell nanofluid can be written as (Buongiorno[26], Motsa et al.,[27])

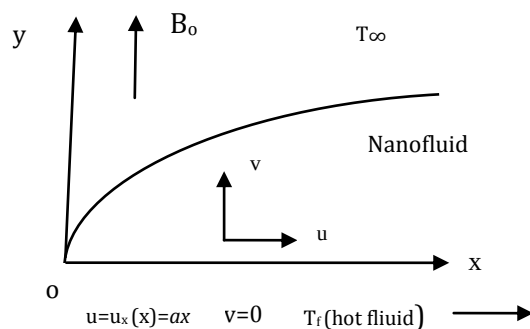


Fig1. Physical model and coordinate system

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + k_0(u^2 \frac{\partial^2 u}{\partial x^2} + v^2 \frac{\partial^2 u}{\partial y^2} + 2uv \frac{\partial^2 u}{\partial x \partial y}) = \nu \frac{\partial^2 u}{\partial y^2} - \frac{\sigma B_0^2}{\rho_f} (u + k_0 v \frac{\partial u}{\partial y}) \quad (2)$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2} + \tau \left\{ D_B \left(\frac{\partial C}{\partial y} \frac{\partial T}{\partial y} \right) + \frac{D_T}{T_\infty} \left(\frac{\partial T}{\partial y} \right)^2 \right\} + \frac{Q_0}{\rho c_p} (T - T_\infty) \quad (3)$$

$$u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} = D_B \frac{\partial^2 C}{\partial y^2} + \frac{D_T}{T_\infty} \left(\frac{\partial^2 T}{\partial y^2} \right) - k_1 (C - C_\infty) \quad (4)$$

Subject to the boundary conditions

$$u = u_w(x) + L \left(\frac{\partial u}{\partial y} \right), v = 0, -k \frac{\partial T}{\partial y} = h_f (T_f - T), C = C_w \text{ at } y = 0$$

$$u = 0, T = T_\infty, C = C_\infty \text{ as } y \rightarrow \infty \quad (5)$$

where u and v are the velocity components along the x and y axes, respectively, ρ_f and ν are the density and kinematic viscosity of the base fluid, σ is the electrical conductivity, $\alpha = k/(\rho C)_f$ is the thermal diffusivity of the base fluid, $\tau = (\rho C)_p/(\rho C)_f$ is the ratio of the nanoparticle heat capacity to the base fluid heat capacity, D_B is the Brownian diffusion coefficient, D_T is the thermophoretic diffusion coefficient, T is the fluid temperature, T_∞ is the ambient fluid temperature, k_0 is the relaxation time of the UCM fluid, k_1 is the rate of chemical reaction, Q_0 is the dimensional heat generation/absorption coefficient, C_p is the specific heat at the constant pressure, L is the hydrodynamic slip factor and C is the nanoparticle concentration. The nanoparticle Concentration at the surface is uniform C_w , which is higher than the ambient fluid concentration C_∞ . The following non-dimensional variables are introduced:

$$\eta = \sqrt{\frac{a}{\nu}} y, \quad \psi(x, y) = \sqrt{a\nu} x f(\eta), \quad \theta(\eta) = \frac{T - T_\infty}{T_f - T_\infty}, \quad \phi(\eta) = \frac{C - C_\infty}{C_w - C_\infty} \quad (6)$$

Where $\psi(x, y)$ is the stream function defined by

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x} \quad (7)$$

which automatically satisfies continuity Eq. (1). In view of Eq. (6), Eqs. (2)–(5) become

$$f''' - M^2 f' + (1 + \lambda M^2) f f'' - f'^2 - \lambda(f^2 f''' - 2 f f' f'') = 0 \quad (8)$$

$$\theta'' + Pr(f\theta' + \delta\theta + Nb\phi'\theta' + Nt\theta'^2) = 0 \quad (9)$$

$$\phi'' + Le f\phi' + \frac{Nt}{Nb}\theta'' - \gamma Le\phi = 0 \quad (10)$$

With boundary conditions being

$$f = S, \quad f' = 1 + S_v f''(0), \quad \theta' = -Bi(1 - \theta), \quad \phi = 1 \quad \text{at } \eta = 0, \\ f' \rightarrow 0, \quad \theta \rightarrow 0, \quad \phi \rightarrow 0 \quad \text{as } \eta \rightarrow \infty \quad (11)$$

Here, the prime denotes differentiation with respect to η , f is the similarity function, θ is the dimensionless temperature, ϕ is the dimensionless nanoparticle concentration,

$Pr = \nu/\alpha$ is the Prandtl number, $Le = \nu/D_B$ is the Lewis number, S_v is the momentum slip, $Bi = (h_f/k)\sqrt{\nu/a}$ is the Biot number, $\lambda = ak_0$ is the elastic parameter, $\delta = Q_0/(\rho C_p a)$ is the

heat generation/absorption parameter $Bi = \left(\frac{h_f}{k} \right) \sqrt{\frac{v}{a}}$ is the magnetic parameter,

$Nb = \frac{((\rho c)_p D_B (C_w - C_\infty))}{((\rho c)_f v)}$ is the Brownian motion parameter, and

$Nt = \frac{((\rho c)_p D_T (T_f - T_\infty))}{((\rho c)_f v T_\infty)}$ is the thermophoresis parameter. It should be mentioned

that $\gamma = k_1 a$, $\gamma > 0$ indicates a destructive chemical reaction while $\gamma < 0$ corresponds to a generative chemical reaction. The case $\delta > 0$ corresponds to internal heat generation while $\delta < 0$ corresponds to internal absorption. The quantities of physical interest in this problem are the skin friction coefficient C_f , the local Nusselt number Nu_x and the local Sherwood number Sh_x :

$$C_f = \frac{\tau_w}{\rho u_w^2}, \quad Nu_x = \frac{x q_w}{k(T_f - T_\infty)}, \quad Sh_x = \frac{x h_m}{D_B(C_w - C_\infty)} \quad (12)$$

Where τ_w is the shear stress, q_w and h_m are the surface heat and mass flux, respectively:

$$\tau_w = \mu(1 + \lambda) \left(\frac{\partial u}{\partial y} \right)_{y=0}, \quad q_w = -k \left(\frac{\partial T}{\partial y} \right)_{y=0}, \quad h_m = -D_B \left(\frac{\partial C}{\partial y} \right)_{y=0} \quad (13)$$

The dimensionless forms of skin friction, the local Nusselt number, and the local Sherwood number become

$$Re_x^{1/2} C_f = (1 + \lambda) f''(0), \quad \frac{Nu_x}{Re_x^{1/2}} = -\theta'(0), \quad \frac{Sh_x}{Re_x^{1/2}} = -\phi'(0)$$

Where $Re_x^{1/2} = \frac{x u_w}{\nu}$ is the local Reynolds number.

Method Of Solution:

The set of non-linear ordinary differential equations (8) – (10) with boundary conditions (11) have been solved by using the Runge - Kutta fourth order method along with shooting. technique. The step size $\Delta \eta = 0.01$ is used while obtaining the numerical solution with $\eta_{\max}$, and accuracy to the fifth decimal place is sufficient for convergence. A step size of $\Delta \eta = 0.01$ was selected to be satisfactory for a convergence criterion of 10^6 in nearly all cases. The maximum value of η_∞ to each group of parameters $M, \lambda, \delta, Pr, Nt, Nb, Le, Bi, S_f$ and γ determined when the values of unknown boundary conditions at $\eta = 0$ not change to successful loop with error less than 10^6 .

Numerical estimates and related discussion:

The set of nonlinear ordinary differential Eqs. (8)–(10) subject to the boundary conditions Eq. (11) were numerically solved using the shooting method with the fourth-order Runge–Kutta method for various values of the magnetic parameter M , elastic parameter λ , thermophoresis parameter N_t , Brownian motion parameter N_b , chemical reaction parameter γ , Biot number Bi , Prandtl number Pr , Lewis number Le , slip velocity S_f and heat generation and absorption δ . The values of the local skin friction coefficient C_f (in terms of the wall velocity gradient $f''(0)$), local Nusselt number Nu_x (in terms of heat transfer rate $-\theta'(0)$) and local Sherwood number Sh_x (in terms of mass transfer rate $-\phi'(0)$) are given in Tables 3 and 4.

From Table 3, the magnitude of C_f and the values of Sh_x increase with increasing λ , whereas the opposite trend is seen for the values of Nu_x . Physically, a high elastic parameter corresponds to solid like behavior and a low elastic parameter corresponds to liquid-like behavior. This phenomenon leads to enhance both the magnitude of C_f and the value of Sh_x . The magnitude of C_f increases with increasing M . This is because the presence of a magnetic field leads to a decrease of the velocity, which increases the velocity gradient at the stretching sheet. The values of Nu_x decrease with increasing M , whereas the opposite trend is observed for the values of Sh_x . From Table 4, the values of Nu_x and Sh_x increase with γ for destructive chemical reaction ($\gamma > 0$) whereas the opposite trend is seen for generative chemical reaction ($\gamma < 0$). Nu_x increases with δ for the heat absorption case ($\delta < 0$) whereas the opposite trend is seen for the heat generation case ($\delta > 0$). For heat generation ($\delta > 0$) the values of Sh_x increase with δ , whereas the opposite trend is observed for heat absorption ($\delta < 0$). Additionally, Nu_x and Sh_x increase with increasing Bi .

The influences of all of the previous parameters on the velocity, temperature, and nanoparticle concentration distributions were investigated, and the results are graphically presented in Figs. 2–13. Figs. 2 and 3 show that the influence of magnetic parameter M and elastic parameter λ on the velocity distribution. From Fig. 2, the velocity distributions and the boundary layer thickness decrease with increasing M . This is because addition of a transverse magnetic field to an electrically conducting nanofluid will give rise to a resistive type of force known as the Lorenz force. This force has the tendency to slow down the velocity of the fluid in the boundary layer. From Fig. 3, the velocity distributions and the boundary layer thickness decrease with increasing λ . physically, the elastic parameter depends on the relaxation time. A larger relaxation time decelerates the motion of the fluid. This helps to decrease the velocity field and the boundary layer thickness.

Fig. 4 and 5 shows the influence of Nt on the temperature and nanoparticle concentration distributions. It is observed that the temperature and nanoparticle concentration distributions are enhanced with increasing Nt . This is because the thermophoretic force generated by the temperature gradient results in fast flow away from the stretching surface. This serves to enhance both the thermal boundary layer thickness and concentration boundary layer thickness. Fig. 6 and 7 shows the influence of Nb on the temperature and nanoparticle concentration distributions. Plot 6. shows the temperature and thermal boundary layer thickness increase with increasing Nb . This is because with increasing Nb , the random motion of the nanoparticles increases resulting in augmentation of the temperature field. From Fig. 7, the nanoparticle concentration distributions decrease with increasing Nb near the surface, whereas the opposite trend occurs far away from the surface.

Figure 8 demonstrates the effect of Prandtl number on the temperature distribution. The numerical results show that an increase in the Prandtl number results in a decrease of the thermal boundary layer thickness and in general lower average temperature within the boundary layer. The reason is that smaller values of Pr are equivalent to increasing the thermal conductivity of the fluid, and therefore heat is able to diffuse away from the heated surface more rapidly than higher values of Pr .

Hence in the case of smaller Pr , the thermal boundary layer is thicker and the rate of heat transfer is reduced. Therefore an increase in Pr reduces temperature.

Fig. 9 shows the influence of Bi on the temperature distributions. From Fig.9, the temperature distribution increase with increasing Bi . The Biot number enhances only the surface temperature because $\theta' = -Bi(1-\theta)$ (Eq. (11)). Increasing the Biot number leads to an increase in the surface temperature, which increases the temperature and the thermal boundary layer thickness. These results are identical to those reported by Makinde and Aziz (3). $Bi \rightarrow \infty$ gives the isothermal surface ($\theta(0) = 1$), which is discussed by Khan and Pop (28).

The variations in the temperature profiles for changes in the heat source/sink are presented in Fig. 10. It is observed that the temperature enhances with an increase in heat source. Figure 11 presents the velocity profiles for various values of velocity-slip factor. With the increasing values of velocity slip parameter the fluid velocity reduces. When slip occurs, the fluid velocity near the plate is no longer equal to the sheet stretching velocity, i.e., a velocity slip exists. With the rise in S_v such a slip velocity decreases. In the case of the no-slip condition, the fluid velocity adjacent to a solid surface is equal to the velocity of the stretching sheet *i.e.*, $f'(0)=1$ while with an increase in other slip parameters, the momentum boundary layer thickness decreases.

Fig. 12 demonstrates the influence of Le on the nanoparticle concentration distributions. From Fig. 12, increasing Le decreases the nanoparticle concentration distributions. The Lewis number is the ratio of the momentum diffusivity to the Brownian diffusion coefficient. Physically, increasing the Lewis number decreases the Brownian diffusion coefficient. This result decreases the concentration field and concentration boundary layer thickness.

Fig. 13 is aimed to shed light on the effect of γ the chemical reaction parameter on the nanoparticle concentration distributions. From Fig. 13, the concentration distributions decrease with increasing γ . The nanoparticle concentration and concentration boundary layer thickness increase with γ magnitude for $\gamma < 0$ case, whereas the opposite trend is observed for $\gamma > 0$ case. Physically, the presence of a destructive reaction ($\gamma > 0$) causes transformation of the species, which decreases the concentration boundary layer thickness. $\gamma < 0$ indicates a generative chemical reaction (i.e., the species diffuses from the stretching sheet in the free stream). Hence, the concentration boundary layer thickness increases with a generative chemical reaction. The destructive chemical reaction ($\gamma > 0$) causes heat production (i.e., thermal energy) in the boundary layer. This helps to enhance the thermal boundary layer. The opposite effect is observed with a generative chemical reaction ($\gamma < 0$). Hence, the temperature of the fluid decreases.

Table:1 Comparison of Nu_x and Sh_x for various values of Nt with $\lambda = 0.1$, $M = 0.5$, $\gamma = 0.22$, $\delta = 0$, $Bi = 1$, $Pr = 0.72$, $Le = 1$, and $Nb = 0.1$

Nt	Makinde and Aziz(2011)		present study	
	$-\theta'(0)$	$-\phi'(0)$	$-\theta'(0)$	$-\phi'(0)$
0.1	0.0929	2.2774	0.0929	2.2726
0.2	0.0927	2.249	0.0927	2.2439
0.3	0.0925	2.2228	0.0925	2.2174
0.4	0.0923	2.1992	0.0923	2.1935
0.5	0.0921	2.1783	0.0921	2.1723

Table:2 Comparison of Nu_x and Sh_x for various values of Nb with

$\lambda = 0.1, M = 0.5, \gamma = 0.22, \delta = 0, Bi = 1, Pr = 0.72, Le = 1$, and $Nt = 0.1$

Nb	Makinde and Aziz(2011)		present study	
	$-\theta'(0)$	$-\phi'(0)$	$-\theta'(0)$	$-\phi'(0)$
0.1	0.0929	2.2774	0.0929	2.2726
0.2	0.0873	2.3109	0.0873	2.3062
0.3	0.0769	2.3299	0.07685	2.3251
0.4	0.0597	2.3458	0.0596	2.3408
0.5	0.0383	2.356	0.0383	2.3509

Table:3 Cf , Nu_x , and Sh_x for various values of λ and M with $Pr = Le = \gamma = Bi = 1$,

$Nb = 0.1, Nt = 0.5$ and $\delta = 0.5$

λ	M	$(1+\lambda)f''$	$-\theta'$	$-\phi'$
0.5	0.0	-1.0081	2.4218	1.3816
0.5	0.1	-1.0091	2.3741	1.3849
0.5	0.2	-1.0106	2.346	1.4093
0.5	0.3	-1.1774	2.2955	2.9437
0.0	0.1	-1.0190	2.2568	-2.6312
1.0	0.1	-1.2502	2.2015	3.1388

Table: 4 Nu_x and Sh_x for various values of γ , Bi , and δ with $Pr = Le = 1, Nb = 0.1$,

$Nt = 0.5, M = 0.1$, and $\lambda = 1$.

γ	Bi	δ	$-\theta'(0)$	$-\phi'(0)$
1.0	1.0	-0.5	0.45586	0.23286
1.0	1.0	0.0	0.33403	0.86948
1.0	1.0	0.5	0.19367	1.47162
1.0	2.0	0.5	0.26199	1.78782
-0.5	1.0	0.5	0.21807	3.34453

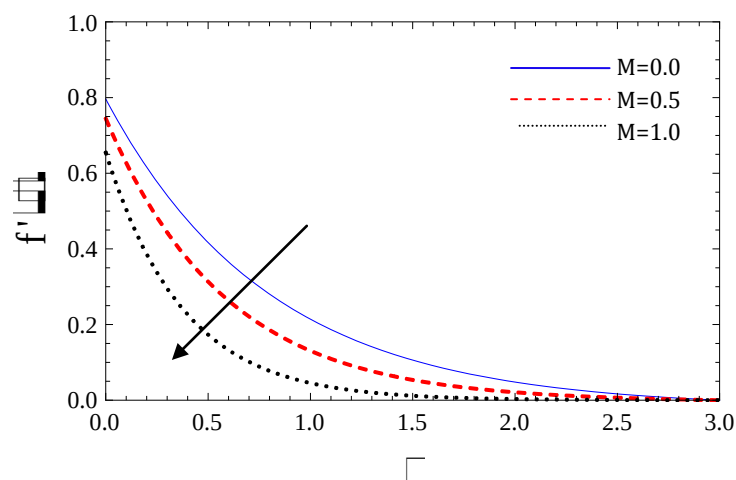


Fig 2. Effects of magnetic field parameter M on velocity profile.

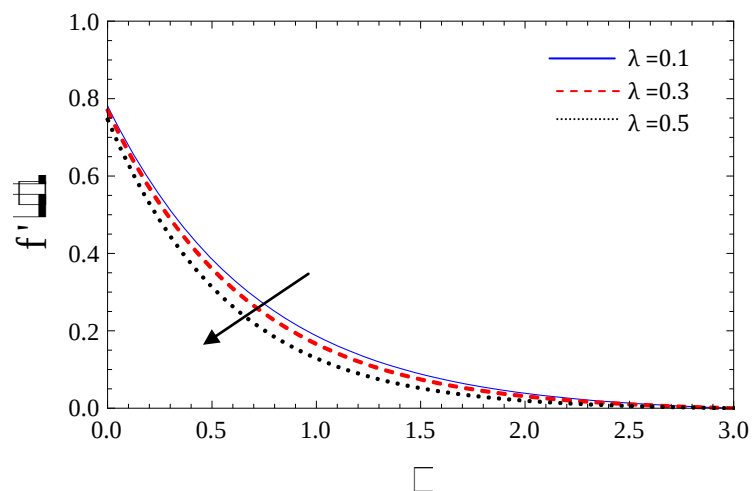


Fig 3. Effects of various values of elastic parameter λ on velocity profile.

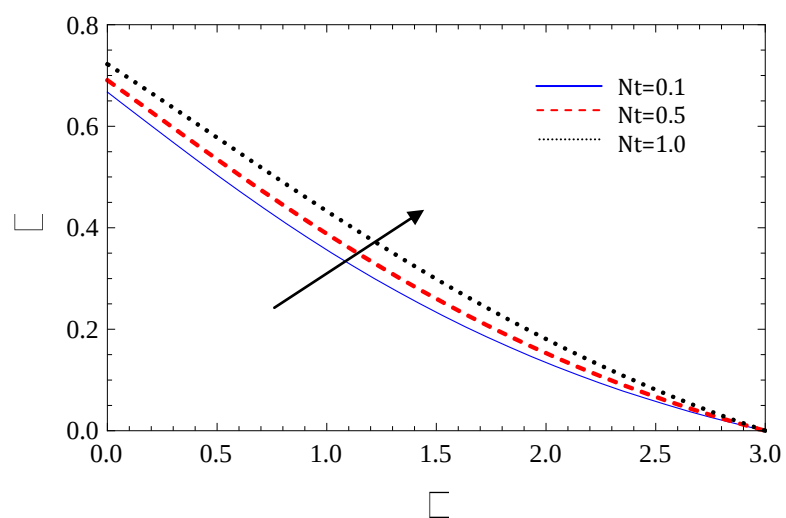


Fig 4. Effects of various values of Nt on temperature profile

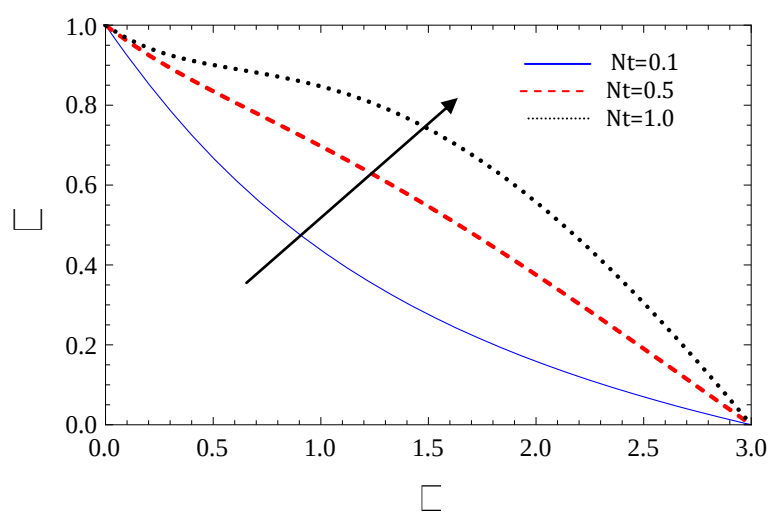


Fig 5. Effects of various values of Nt on Concentration profile

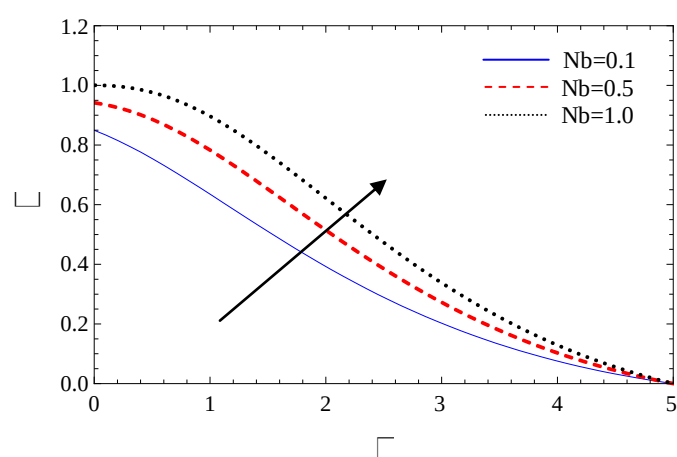


Fig 6. Effects of various values of Nb on temperature profile

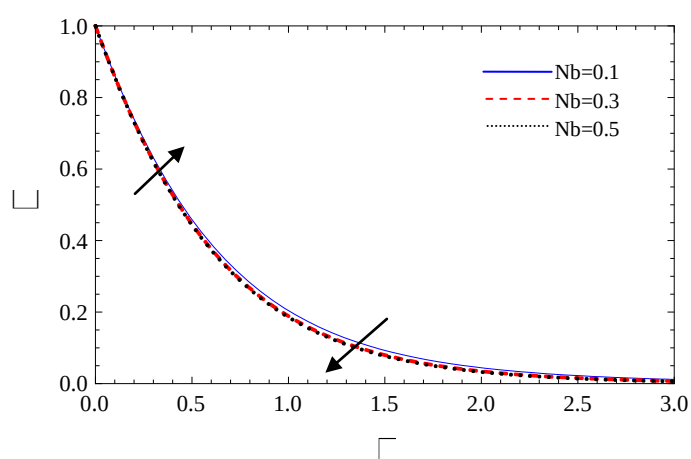


Fig 7. Effects of various values of Nb on Concentration profile

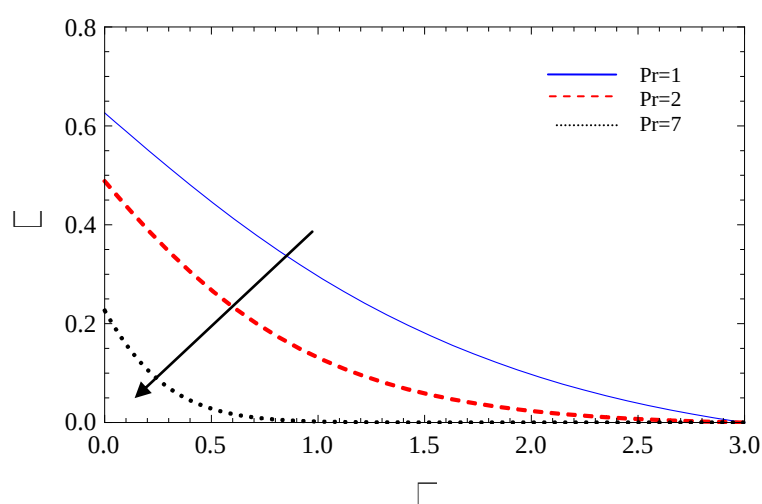


Fig 8. Effects of various values of Pr on Temperature profile

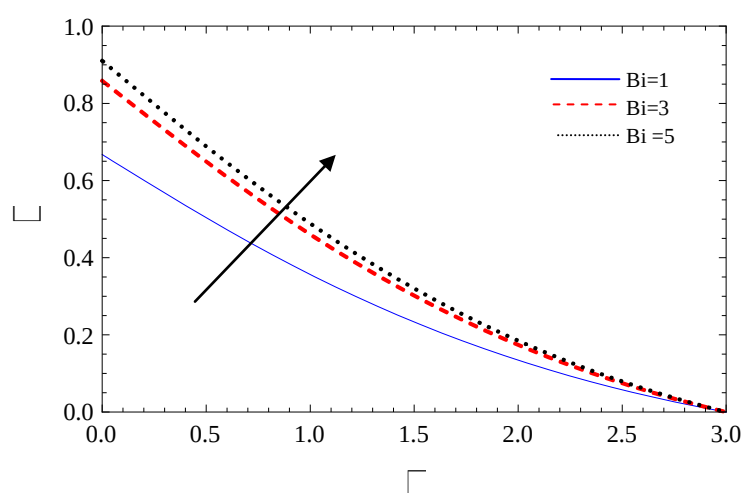


Fig 9. Effects of various values of Bi on Temperature profile

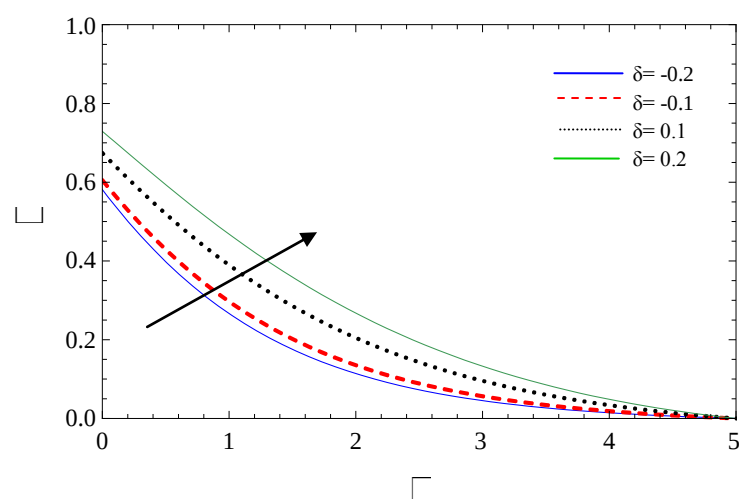


Fig 10. Effects of various values of δ on Temperature profile

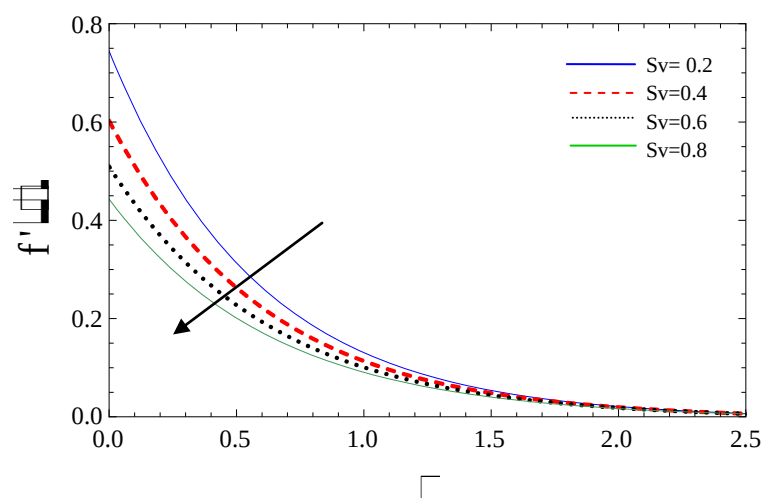


Fig 11. Effects of various values of S_v on velocity profile

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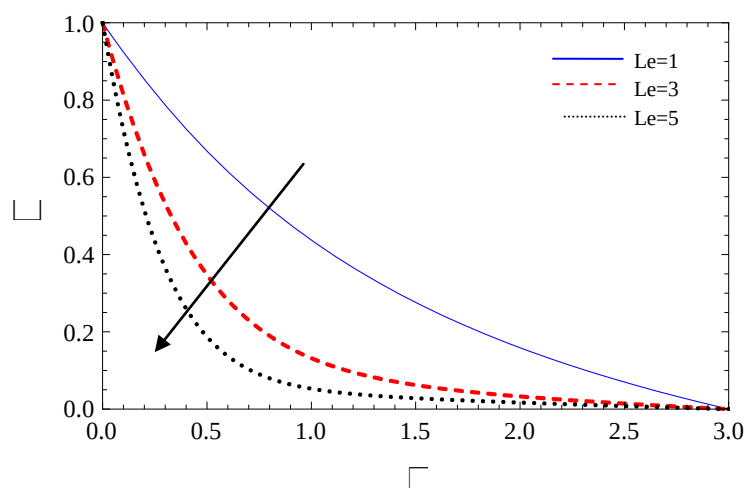


Fig 12. Effects of various values of Le on Concentration profile

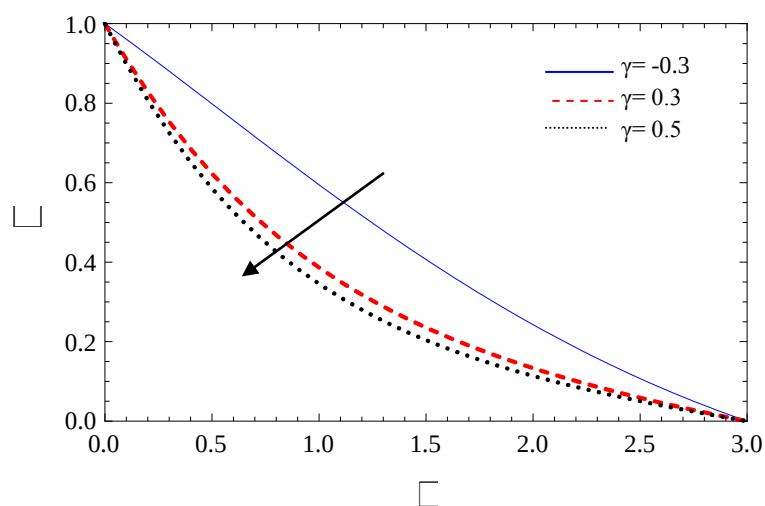


Fig 13. Effects of various values of γ on Concentration profile

Conclusions:

A two-dimensional boundary layer flow of a Maxwell nanofluid over a stretching surface with a convective boundary condition is considered. Chemical reaction, MHD and velocity slip effects were also numerically investigated. The governing equations were reduced to a system of ordinary differential equations and then the equations were solved. The validity of the present analysis was established by comparing the existing results with previously published data. The main results of the present analysis are as follows:

1. The magnitude of the skin friction coefficient and local Sherwood number increase, whereas the local Nusselt number decreases with increasing λ and M .
2. The local Nusselt number and local Sherwood number increase with increasing Bi number.
3. The local Nusselt number and local Sherwood number increase with γ for destructive chemical reaction ($\gamma > 0$), whereas the opposite trend is observed for generative chemical reaction ($\gamma < 0$).
4. The local Nusselt number increases with δ for heat absorption ($\delta < 0$), whereas the opposite trend occurs for heat generation ($\delta > 0$).
5. The local Sherwood number increases with δ for $\delta > 0$, whereas the opposite trend is observed for $\delta < 0$.
6. The velocity distributions decrease whereas the temperature and nanoparticle concentration distributions increase with increasing M and λ .
7. The concentration distributions decrease with increasing γ , whereas the opposite trend is observed for the temperature distributions.
8. With the increasing values of velocity slip parameter the fluid velocity reduces.

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